niram-css v0.2.0

The `niram-css` package lets you use standard CSS color names directly within your Typst documents. The word *niram* (நிறம்) means "color" in Tamil, reflecting the package's singular focus on color.

While Typst includes a curated set of predefined colors, accessing the full spectrum of familiar CSS keywords often requires hunting down exact hex or RGB values. `niram-css` eliminates this friction by bringing all 147 standardized CSS colors natively into your workflow. Instead of pausing to look up a value like `#6495ed`, you can simply type `cornflowerblue` and let the package do the rest.

Key features include:

- **Complete CSS Color Support:** Access all 147 standard color keywords from the CSS3 specification (see Table 1, or view them in a browser).
- **Flexible Naming Conventions:** Write color names in the format most natural to you. The package automatically normalizes inputs, meaning `cornflowerblue`, `cornflower blue`, `cornflower-blue`, `cornflower_blue`, and `CornflowerBlue` all resolve to the exact same color.

Requirements

This package requires Typst 0.13.0 or higher.

Getting Started

Import the package at the top of your Typst document:

```
#import "@preview/niram-css:0.2.0": *
```

Then, use the `css` function anywhere a color value is expected:

```
Normal text may be #text(fill: css("crimson"))[colored so.]
```

Normal text may be colored so.

That is all it takes to start using CSS color names in your document.

Usage

Function Reference

```
css(color-name: str) -> color
```

color-name `str`

A string representing the color name. The following conventions are all accepted and normalized to a canonical form before lookup:

Style	Example	Common in
No separator	`cornflowerblue`	CSS/SVG
Spaces	`cornflower blue`	X11/GIMP/Matplotlib
Hyphens	`cornflower-blue`	CSS tooling/LaTeX xcolor
Underscores	`cornflower_blue`	X11 alternate/Python/R
CamelCase	`CornflowerBlue`	X11/.NET/Java

Color name matching is completely case-insensitive; `AliceBlue`, `aliceblue`, and `ALICEBLUE` are all treated identically.

-> `color`

The `css` function returns an RGB color. Because the returned value is a standard Typst color type, you can seamlessly modify it using any of Typst's built-in color functions, such as `lighten` or `darken`.

Examples

The examples below assume that `#import "@preview/niram-css:0.2.0": *` is at the top of the document.

1. Filling a rectangle with a CSS color.

```
#rect(width: 100%, height: auto, fill: css("darkorchid"))
```

2. CSS color names can be written in any supported format. Lowercase and CamelCase are shown here.

```
#grid(
  columns: 2,
  rows: 2,
  row-gutter: 10pt,
  column-gutter: 10pt,
  [(a) saddlebrown],
  [#box(width: 30mm, height: 1em, fill: css("saddlebrown"))],
  [(b) SaddleBrown],
  [#box(width: 30mm, height: 1em, fill: css("SaddleBrown"))],
)
```

(a) saddlebrown

(b) SaddleBrown

3. Typst's built-in `maroon` has a different hex value than the standard CSS `maroon`. To highlight this distinction, the CSS version in the example below uses an initial uppercase letter (`Maroon`).

```
#grid(
  columns: 3,
  rows: 2,
  row-gutter: 10pt,
  column-gutter: 10pt,
  [Typst maroon],
  [#color.maroon.to-hex()],
  [#box(width: 30mm, height: 1em, fill: maroon)],
  [CSS maroon],
  [#css("Maroon").to-hex()],
  [#box(width: 30mm, height: 1em, fill: css("Maroon"))],
)
```

Typst maroon	#85144b	
CSS maroon	#800000	

4. Using CSS colors to underline and double-underline text.

```
Thin #underline(stroke: css("darkblue"), offset: 2pt)[underline.]

Double #underline(
  stroke: (paint: css("darkblue"), thickness: 0.5pt),
  offset: 2pt,
)[
  #underline(
    stroke: (paint: css("darkblue"), thickness: 0.5pt),
    offset: 5pt,
  )[underline.]
]

Thick #underline(stroke: (paint: css("darkblue"), thickness: 2pt), offset: 2pt)
[underline.]
```

Thin underline.

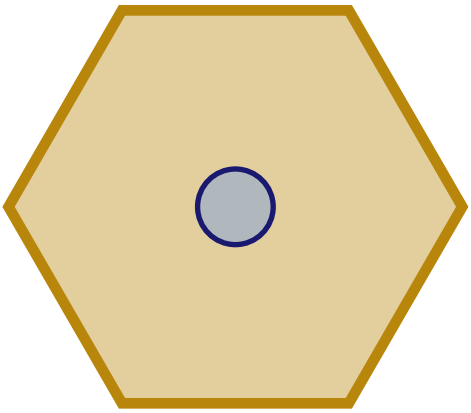
Double underline.

Thick underline.

5. Using CSS colors in drawing and data visualization.

```
#align(center)[
  #block[
    #polygon.regular(
      fill: css("darkgoldenrod").lighten(60%),
      stroke: (paint: css("darkgoldenrod"), thickness: 4pt),
      size: 60mm,
      vertices: 6,
    )
    #place(center + horizon)[
      #circle(
        radius: 5mm,
```

```
        stroke: (paint: css("midnightblue"), thickness: 2pt),
        fill: css("cornflowerblue").transparentize(60%),
      )
    ]
  ]
]
```



The History of the CSS Color Standard

X11 Colors

The X11 color names [1], developed in the 1980s, included a file named `rgb.txt` that was shipped with every X11 installation. It mapped over 400 color names to specific RGB values. This list was used in Unix-based systems to standardize color representation, but was never formally standardized across web formats, leading to variations between distributions.

Sixteen Named Colors

The HTML 3.2 specification [2] of 1996 introduced sixteen color names, derived from the standard sixteen colors supported by the Windows VGA palette. These colors, called the sixteen named colors, are:

- `aqua`
- `black`
- `blue`
- `fuchsia`
- `gray`
- `green`
- `lime`
- `maroon`
- `navy`
- `olive`
- `purple`
- `red`
- `silver`
- `teal`
- `white`

- `yellow`

Thanks to this standard, these sixteen color names could be used reliably and consistently in documents across the Web, without needing to specify hexadecimal values.

The CSS Level 1 colors introduced in 1996 specified these same sixteen colors, while the CSS Level 2 colors, introduced in 1998, expanded to seventeen colors with the addition of `orange`.

CSS/SVG Colors

Finally, the SVG 1.1 Specification of 2011 [3] and the CSS Color Module Level 3 [4] of 2022 standardized on a list of 147 color names, known as the "recognized color keyword names." They comprise the original HTML/VGA named colors, `orange`, and additional colors drawn from the X11 set.

This standardized list provides a consistent color naming scheme across the Web and has remained stable since its adoption. This set of color names forms the basis of this package. How the CSS colors were given their rather idiosyncratic names is explained in a factual but lighthearted fashion by Alex Sexton in this YouTube video [5].

While all 147 CSS color names are distinct, only 138 have unique hex values. The nine pairs of color names that share identical hex values are detailed in Table 2.

Typst's Predefined Colors versus CSS Colors

Typst ships with eighteen predefined colors, including the sixteen HTML named colors, `orange`, and one additional color, `eastern`, which has no CSS equivalent.

The remaining seventeen Typst colors share names with their CSS counterparts, but—with the exception of `white` and `black`—they have different hex values, as shown in Table 3.

X11 Gray Equivalents

X11 color coverage is intentionally out of scope for this package. The X11 database includes numbered color families (e.g. `red1–red4`, `blue1–blue4`, `gray0–gray100`) that are rarely used today.

However, any X11 numbered gray shade can be reproduced exactly with Typst's `lighten` function as demonstrated in Table 4.

CSS Named Color Reference

Color Name	Hex Value	Swatch
aliceblue	#f0f8ff	
antiquewhite	#faebd7	
aqua	#00ffff	
aquamarine	#7fffd4	
azure	#f0ffff	
beige	#f5f5dc	
bisque	#ffe4c4	
black	#000000	
blanchedalmond	#ffebcd	
blue	#0000ff	
blueviolet	#8a2be2	
brown	#a52a2a	
burlywood	#deb887	

Color Name	Hex Value	Swatch
cadetblue	#5f9ea0	
chartreuse	#7fff00	
chocolate	#d2691e	
coral	#ff7f50	
cornflowerblue	#6495ed	
cornsilk	#fff8dc	
crimson	#dc143c	
cyan	#00ffff	
darkblue	#00008b	
darkcyan	#008b8b	
darkgoldenrod	#b8860b	
darkgray	#a9a9a9	
darkgreen	#006400	

Color Name	Hex Value	Swatch
darkgrey	#a9a9a9	
darkkhaki	#bdb76b	
darkmagenta	#8b008b	
darkolivegreen	#556b2f	
darkorange	#ff8c00	
darkorchid	#9932cc	
darkred	#8b0000	
darksalmon	#e9967a	
darkseagreen	#8fbc8f	
darkslateblue	#483d8b	
darkslategray	#2f4f4f	
darkslategrey	#2f4f4f	
darkturquoise	#00ced1	

Color Name	Hex Value	Swatch
darkviolet	#9400d3	
deeppink	#ff1493	
deepskyblue	#00bfff	
dimgray	#696969	
dimgrey	#696969	
dodgerblue	#1e90ff	
firebrick	#b22222	
floralwhite	#fffaf0	
forestgreen	#228b22	
fuchsia	#ff00ff	
gainsboro	#dcdcdc	
ghostwhite	#f8f8ff	
gold	#ffd700	

Color Name	Hex Value	Swatch
goldenrod	#daa520	
gray	#808080	
green	#008000	
greenyellow	#adff2f	
grey	#808080	
honeydew	#f0fff0	
hotpink	#ff69b4	
indianred	#cd5c5c	
indigo	#4b0082	
ivory	#fffff0	
khaki	#f0e68c	
lavender	#e6e6fa	
lavenderblush	#fff0f5	

Color Name	Hex Value	Swatch
lawngreen	#7cfc00	
lemonchiffon	#fffacd	
lightblue	#add8e6	
lightcoral	#f08080	
lightcyan	#e0ffff	
lightgoldenrodyellow	#fafad2	
lightgray	#d3d3d3	
lightgreen	#90ee90	
lightgrey	#d3d3d3	
lightpink	#ffb6c1	
lightsalmon	#ffa07a	
lightseagreen	#20b2aa	
lightskyblue	#87cefa	

Color Name	Hex Value	Swatch
lightslategray	#778899	
lightslategrey	#778899	
lightsteelblue	#b0c4de	
lightyellow	#ffffe0	
lime	#00ff00	
limegreen	#32cd32	
linen	#faf0e6	
magenta	#ff00ff	
maroon	#800000	
mediumaquamarine	#66cdaa	
mediumblue	#0000cd	
mediumorchid	#ba55d3	
mediumpurple	#9370db	

Color Name	Hex Value	Swatch
mediumseagreen	#3cb371	
mediumslateblue	#7b68ee	
mediumspringgreen	#00fa9a	
mediumturquoise	#48d1cc	
mediumvioletred	#c71585	
midnightblue	#191970	
mintcream	#f5fffa	
mistyrose	#ffe4e1	
moccasin	#ffe4b5	
navajowhite	#ffdead	
navy	#000080	
oldlace	#fdf5e6	
olive	#808000	

Color Name	Hex Value	Swatch
olivedrab	#6b8e23	
orange	#ffa500	
orangered	#ff4500	
orchid	#da70d6	
palegoldenrod	#eee8aa	
palegreen	#98fb98	
paleturquoise	#afeeee	
palevioletred	#db7093	
papayawhip	#ffefd5	
peachpuff	#ffdab9	
peru	#cd853f	
pink	#ffc0cb	
plum	#dda0dd	

Color Name	Hex Value	Swatch
powderblue	#b0e0e6	
purple	#800080	
red	#ff0000	
rosybrown	#bc8f8f	
royalblue	#4169e1	
saddlebrown	#8b4513	
salmon	#fa8072	
sandybrown	#f4a460	
seagreen	#2e8b57	
seashell	#fff5ee	
sienna	#a0522d	
silver	#c0c0c0	
skyblue	#87ceeb	

Color Name	Hex Value	Swatch
slateblue	#6a5acd	
slategray	#708090	
slategrey	#708090	
snow	#fffafa	
springgreen	#00ff7f	
steelblue	#4682b4	
tan	#d2b48c	
teal	#008080	
thistle	#d8bfd8	
tomato	#ff6347	
turquoise	#40e0d0	
violet	#ee82ee	
wheat	#f5deb3	

Color Name	Hex Value	Swatch
`white`	`#ffffff`	
`whitesmoke`	`#f5f5f5`	
`yellow`	`#ffff00`	
`yellowgreen`	`#9acd32`	

Table 1: All 147 CSS named colors with their hex values and swatches.

CSS Colors Sharing the Same Hex Values

Color Names	Hex Value	Swatch
aqua cyan	`#00ffff`	
darkgray darkgrey	`#a9a9a9`	
darkslategray darkslategrey	`#2f4f4f`	
dimgray dimgrey	`#696969`	
fuchsia magenta	`#ff00ff`	
gray grey	`#808080`	
lightgray lightgrey	`#d3d3d3`	
lightslategray lightslategrey	`#778899`	
slategray slategrey	`#708090`	

Table 2: CSS colors sharing the same hex value.

Typst's Predefined Colors versus CSS Colors

Color Name	Typst Hex Value	Swatch	CSS Hex Value	Swatch
aqua	#7fdbff		#00ffff	
black	#000000		#000000	
blue	#0074d9		#0000ff	
eastern	#239dad		—	—
fuchsia	#f012be		#ff00ff	
gray	#aaaaaa		#808080	
green	#2ecc40		#008000	
lime	#01ff70		#00ff00	
maroon	#85144b		#800000	
navy	#001f3f		#000080	
olive	#3d9970		#808000	
orange	#ff851b		#ffa500	
purple	#b10dc9		#800080	

Color Name	Typst Hex Value	Swatch	CSS Hex Value	Swatch
`red`	`#ff4136`		`#ff0000`	
`silver`	`#dddddd`		`#c0c0c0`	
`teal`	`#39cccc`		`#008080`	
`white`	`#ffffff`		`#ffffff`	
`yellow`	`#ffdc00`		`#ffff00`	

Table 3: Typst's predefined colors and their CSS counterparts, with hex values and swatches.

X11 Gray Equivalents

X11 Name	Typst Expression	Hex Value	Swatch
gray0 grey0	black.lighten(0%)	#000000	
gray10 grey10	black.lighten(10%)	#1a1a1a	
gray20 grey20	black.lighten(20%)	#333333	
gray30 grey30	black.lighten(30%)	#4c4c4c	
gray40 grey40	black.lighten(40%)	#666666	
gray50 grey50	black.lighten(50%)	#808080	
gray60 grey60	black.lighten(60%)	#999999	
gray70 grey70	black.lighten(70%)	#b2b2b2	
gray80 grey80	black.lighten(80%)	#cccccc	
gray90 grey90	black.lighten(90%)	#e6e6e6	
gray100 grey100	black.lighten(100%)	#ffffff	

Table 4: X11 numbered gray equivalents using `black.lighten(n%)`.

References

[1] Wikipedia contributors, "X11 color names," 2025. [Online]. Available: https://en.wikipedia.org/w/index.php?title=X11_color_names

[2] W3C, "HTML 3.2 Features At a Glance," Dec. 09, 1996. [Online]. Available: https://www.w3.org/MarkUp/Wilbur/features.html

[3] W3C, "Recognized color keyword names," Aug. 10, 2011. [Online]. Available: https://www.w3.org/TR/SVG11/types.html#ColorKeywords

[4] W3C, "Extended color keywords," Jan. 18, 2022. [Online]. Available: https://www.w3.org/TR/css-color-3/#svg-color

[5] Alex Sexton, *Peachpuffs and Lemonchiffons [CSSConfUS2014]*. Accessed: July 17, 2014. [Online Video]. Available: https://www.youtube.com/watch?v=HmStJQzclHc